Predictive Modeling & Risk Scoring for Bank Customer Churn

Academic Research & Executive Technical Report | Enterprise AI Platform v1.0

1. Executive Abstract

This project implements an end-to-end full-stack machine learning architecture for predicting bank customer churn and computing operational risk scores. Using 10,000 retail banking records across France, Germany, and Spain, five models were benchmarked through 5-fold stratified cross-validation. The winning Gradient Boosting Pipeline achieves an ROC-AUC of 0.8690, PR-AUC of 0.7184, Accuracy of 86.85%, and Brier Score of 0.0994 on an untouched test set of 2,000 customers. At the calibrated decision threshold of 0.28, minority churn recall reaches 67.32% with an F1-score of 64.02%.

2. Data Hygiene & Feature Engineering

Customer identifiers (CustomerId, Surname, RowNumber) were quarantined and excluded from predictive features to prevent target leakage. Domain-engineered features include BalanceSalaryRatio (deposit-to-income balance), EngagementProductInteraction (active membership × product count), and ProductDensityIndicator (NumOfProducts / (Tenure + 1)).

3. Cross-Validation Model Comparison

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Gradient Boosting (Selected)	86.29%	75.12%	48.22%	0.5869	0.8607	0.6924
XGBoost	80.44%	51.13%	73.37%	0.6025	0.8602	0.6938
Random Forest	83.36%	56.95%	67.06%	0.6158	0.8577	0.6835
Logistic Regression	75.96%	43.57%	77.67%	0.5583	0.8308	0.6025
Decision Tree	75.46%	43.02%	79.32%	0.5583	0.8231	0.6073

4. Operational Risk Scoring Framework

Risk scores are calculated as: **Risk Score = Churn Probability × 100**.
• **Low Risk (0 - 24.9%):** Standard engagement. (69.8% of portfolio)
• **Medium Risk (25.0 - 49.9%):** Proactive relationship management review. (18.1% of portfolio)
• **High Risk (≥ 50.0%):** Executive intervention and customized retention terms. (12.1% of portfolio)

5. Model Explainability & Partial Dependence (PDP)

Global feature attribution is governed by TreeSHAP and Partial Dependence Plots (PDP):
• **Age:** Pronounced non-linear escalation between ages 40 and 60, where marginal churn probability climbs from ~10.5% to >45%.
• **NumOfProducts:** Non-linear U-curve; 2 products yield optimal customer retention (~11.3% risk), whereas 3 or 4 products surge to ~76.0% and ~86.6%.
• **IsActiveMember:** Active engagement provides an average marginal risk reduction of ~10.3 percentage points (~16.1% vs ~26.4%).

• **Balance:** Modest positive gradient, rising from ~20.6% for zero balance to ~25%–28% for deposits exceeding €100,000.

6. Responsible Use & Governance

Predictions represent statistical estimates derived from historical patterns. The model is designated as a decision-support instrument and must not be used to automatically restrict credit, deny banking services, or enforce punitive pricing without human review.